

Genetic dissection of early-season cold tolerance in sorghum (*Sorghum bicolor* (L.) Moench)

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Abstract Soil temperatures at 15°C or below limit germination and seedling establishment for warm season cereal crops such as sorghum (*Sorghum bicolor* (L.) Moench) during early-season planting. To better understand the genetics of early-season cold tolerance in sorghum, mapping of quantitative trait loci (QTL) associated with germination, emergence and vigor using a recombinant inbred mapping population was carried out. A mapping population consisting of 171 F₇–F₈ recombinant inbred lines (RILs) derived from the cross between RTx430 (cold-sensitive) and PI610727 (cold-tolerant) was developed and a genetic map was constructed using 141 microsatellites or simple sequence repeat (SSR) markers. The RILs were

evaluated for cold and optimal temperature germinability in the laboratory, field emergence, and seedling vigor in two locations during early-season planting. Two or more QTL were detected for all traits, except for seedling vigor, with only one QTL was detected in the population. A QTL for cold germinability (*Germ 12-2.1*) showed the highest LOD value and was also associated with optimal germinability. One of the QTL for field emergence, *Fearlygerm-9.3*, a contribution from PI610727, was found significant in both locations used for the study. This study showed alignment of QTL in SBI1 (*Fearlygerm-1.2* and *FGerm30-1.2*) with previously reported QTL associated with late field emergence identified from a different mapping population. This indicates that PI610727 shares some common loci with other known early-season cold-tolerant sorghum germplasm but also harbors novel QTL that could be useful in introgression of enhanced laboratory germination and early-season field emergence.

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Introduction

Germination and ensuing seedling establishment of most crops are adversely affected by cool soil and air temperatures. However, the effects of cold stress are

more pronounced for warm season cereals such as sorghum and rice. Low-temperature-induced inhibition of germination and emergence in the field is a common problem encountered in crops that originate from warm environments and that were bred to grow in cooler temperate conditions specifically during early-season sowing (Lu et al. 2007). The inability to germinate prevents seedling emergence, resulting in poor stands, and subsequent growth of seedlings is hampered by cool air temperatures.

As an example, in rice, most germplasms are susceptible to low temperatures, but some sources of moderate tolerance have been found in the *japonica* group. A landrace, Italica Livorno, was found to exhibit excellent cold tolerance in terms of germinability at 15°C under laboratory conditions (Fujino et al. 2004). This finding shows that genetic variation for cold tolerance exists and is available to be tapped for breeding purposes. An analysis to dissect the genetic basis of tolerance using a backcross population of Italica Livorno resulted in identification of a major QTL associated with the cold tolerance trait (Fujino et al. 2004). Map-based cloning revealed that this QTL encodes a novel gene for peptide involved in endosperm rupture (Fujino et al. 2008). This peptide was hypothesized to be sensitive to low temperature and thus cold germination is significantly reduced (Fujino et al. 2008). For other crops such as sorghum, however, there is limited information on QTL for early-season cold tolerance.

Sorghum (*Sorghum bicolor* (L.) Moench) is a warm season crop known for its drought tolerance and adaptation to high temperature (Doggett 1988), but like rice it is susceptible to adverse effects of low temperature. Most varieties of sorghum under cultivation are vulnerable to cooler conditions during stand establishment at early-season planting from April to May in many areas of the US sorghum belt. Specifically, stand establishment and early-season vigor of sorghum are adversely affected by air and soil temperatures below 15°C during germination, emergence and early seedling growth (Yu and Tuinstra 2001). Robust cold tolerance is highly advantageous in sorghum to allow the crop to be sown earlier to take advantage of spring moisture. Cold tolerance could also facilitate two cropping cycles (an early-early plus regular season plantings). The trait could also be expected to stabilize and

increase yield by establishing excellent crop stand and maintaining high plant density starting at the critical sowing period.

Variations for cold tolerance have been reported and identified within sorghum germplasm pools. Of special interest as sources of cold tolerance are landraces that have evolved in the temperate regions of China (Lu and Dahlberg 2001). Previous reports have shown that Chinese landraces known as “kaoliangs” exhibited higher seedling emergence and improved seedling vigor under cool conditions compared to select US hybrids and elite inbreds (Franks et al. 2006). However, most of these landraces also harbor poor or undesirable agronomic traits. Thus it is deemed important to dissect the cold tolerance trait and devise an effective and efficient means to transmit the trait to elite parental lines of sorghum.

Previously, a Chinese kaoliang, Shanqui Red, has been reported as a parental source of early-season cold tolerance trait (Knoll and Ejeta 2008). To facilitate transfer of the early-season cold tolerance, Knoll et al. (2008) carried out genetic mapping and identification of genome regions associated with cold tolerance. In this study several QTL for cold germinability and one for seedling vigor with low to moderate contribution to genetic variation were identified from Shanqui Red (Knoll and Ejeta 2008; Knoll et al. 2008). As a result, three markers associated with the QTL have been used to verify introgression of cold tolerance in two populations using Shanqui Red as the common donor for cold tolerance (Knoll et al. 2008).

However, as early-season cold tolerance is a complex trait controlled by a number of genes, it is important to identify various sources of tolerance to effectively combine favorable alleles by pyramiding. This approach could serve as an effective approach to produce elite sorghum lines with stable and robust early-season cold tolerance. Here we report the development and genetic dissection of early-season cold tolerance using recombinant inbred lines from a cross between RTx430 and PI610727, with the latter serving as a new source of early-season cold tolerance. The objectives of this study were to identify molecular markers associated with various traits for early-season cold tolerance and to characterize new germplasm source of early-season cold tolerance.

Materials and methods

Plant materials and phenotypic data collection

The recombinant inbred population developed for this study was derived from a cross between RTx430 and PI610727. RTx430 is a widely adapted inbred and an important pollinator/restorer line of sorghum with medium-sized seeds containing yellow endosperm (Miller 1984). PI610727 (also known as Gaigao Liang), a Chinese kaoliang, is a semi-adapted photoperiod-insensitive landrace selected from previous screening of a group of Chinese germplasm for early-season cold tolerance (Franks et al. 2006). PI610727 has white seeds and does not have testa. Notably RTx430 and PI610727 have similar values for flowering time (59 days for RTx430 and 57 days for PI610727 in Lubbock, Texas, USA) but are quite divergent for plant height. The F_1 hybrid was developed by hand emasculation of RTx430 and subsequent pollination with PI610727. A confirmed F_1 plant was self-pollinated and 200 seeds were advanced to the F_2 generation. Individual F_2 plants were selfed and advanced by single seed descent to the F_6 generation. Seeds from each line were bulked after the F_6 generation. The mapping population consisting of 171 F_7 – F_8 recombinant inbred lines.

Seeds of recombinant inbred lines (RILs) and parents were produced in 2008 under optimal agronomic field conditions at the Cropping Systems Research Laboratory, USDA-ARS farm in Lubbock, Texas. High-quality seeds of each RIL were harvested and subsequently stored under optimal conditions in a seed room with temperature and humidity control to maintain seed quality. All seeds in the studies described in this work were obtained from this seed lot. Parents and RILs were screened for differences in cold and optimal temperature germinability using the procedures recommended by the Association of Official Seed Analysts (Association of Official Seed Analysts (AOSA) 1999). Briefly, 25 seeds were sown in polystyrene Petri dishes lined with filter paper moistened with sterile distilled water. Seeds were allowed to incubate/germinate at a constant 12°C for cold germination and for optimal germination at 30°C for 8 h alternating with 20°C for 16 h, in the dark in separate controlled temperature chambers. Germination was determined visually based on protrusion of radicle to approximately 1 mm length.

Final germination was counted at 4 and 7 days after sowing for optimal temperature and cold germination assays, respectively. Each entry is represented by three replications and each experiment was laid in a completely randomized design.

To determine variation in field emergence, the RILs and parents were sown in 5×1 m plots at the USDA-ARS farm in Lubbock, Texas (33°59'N, 101°89'W; USDA_LBK) and at Texas A&M Agrilife farm at New Deal, Texas (33°69'N, 101°82'W; ND_TAMU). A total of 50 manually selected high-quality seeds were sown on top of well-prepared beds on April 1, 2009 at both locations. Plots were uniformly irrigated using a drip system after sowing. Each RIL entry and parents were represented by three replications. The experiment was laid in a randomized complete block design. The average soil and air temperature during the experimental period were recorded by a HOBO data logger at each location (Electronic Supplementary Material Fig. S1). Emergence was assessed by counting the number of plants that had emerged at 30 days after sowing. Seedling vigor was scored based on a previous rating scale by Maiti et al. (1981); where 1 represents the most vigorous and robust seedlings and 5 represents poor vigor, based on size and physical appearance of the seedlings and not on seedling emergence percentages.

DNA extraction, genotyping and linkage map construction

Leaf samples from bulked plant samples (three plants) of parents and RILs were harvested and used for genomic DNA extraction. Genomic DNA was extracted using a high throughput method previously described (Burow et al. 2009).

Genotyping was conducted through an analysis of simple sequence repeat markers or microsatellites by polymerase chain reaction. For this study, a newly developed set of sorghum microsatellite markers which are collectively referred to as SEAMs were used in addition to previously known markers such as the *Xtxp*, *Xcup*, and *Xgap* series of sorghum SSRs. Briefly, SEAMs markers were represented as *Xsbarslbc* series and primer sequence information are available at http://www.lbk.ars.usda.gov/psgd/sorghum/2009SorghumSEAMs_LBKARS.xls (accessed 15 January 2010).

The *Xtxp*, *Xcup*, and *Xgap* markers were used as anchors for verification of linkage order and comparison with other existing maps of sorghum.

Amplification of microsatellites was carried out by polymerase chain reaction as described in Burow et al. (2009). Products were separated using an ABI 3100 machine (Applied Biosystems) or by agarose gel electrophoresis (3.0% using SFR, Amresco, Solon, Ohio, USA). Fragment sizes were visualized after ethidium bromide staining and gels were documented and archived using ChemImager (Alpha Innotech). The genotype data for each marker was tested by the chi-square test to detect segregation distortion. A total of five markers were excluded from further evaluation as a result of the chi-square analysis.

The construction of linkage map was performed using software MAPMAKER/EXP v.3.0b (Lander et al. 1987). The distances between DNA markers in each linkage group were estimated using the Kosambi mapping function (Kosambi 1944). A minimum LOD score of 3.0 and distance of 30 cM was applied in the construction of the linkage map. Linkage groups were aligned to a previously established map by Menz et al. (2002). The linkage groups were then numbered based on chromosome designations from Kim et al. (2005).

Data analysis and QTL mapping

Statistical analyses of phenotypic data were carried out using Excel and SYSTAT software. Germination data from the optimal and cool temperature analyses were transformed using the arcsin function to improve homogeneity of error variances using the following formula:

$$G' = \arcsin G$$

where *G* is the actual germination value (no. of germinated seeds/total no. of seeds used for sowing). Heritability (*H*) estimates were calculated for all traits analyzed. *H* was calculated with the following formula:

$$H = [(G) - MS(E)]/MS(G)$$

where *MS*(*G*) refers to the mean square of the RILs (the genotypes being compared) and *MS*(*E*) is the mean square for the error.

Identification of QTL for the six traits evaluated was performed using QTL Cartographer v.2.5 (Basten et al. 2003). Analysis was done sequentially, starting with single marker analysis, followed by interval mapping and then with the composite interval mapping (CIM) module. CIM was performed using Model 6, with scanning interval of 2 cM between markers. Significant QTL were identified on the basis of CIM analysis using the experiment-wise threshold level determined by computing 1,000 permutations (Churchill and Doerge 1994) as implemented in QTL Cartographer. QTL effects are expressed relative to the RTx430 allele. Graphic representation of linkage groups and QTL were developed using Map Chart 2.1 (Voorrips 2002).

Results

Observations on phenotype of parents and RILs in seed production environment at Lubbock, Texas

It is recognized that one of the most important factors in cold tolerance studies is the use of high-quality undamaged seeds produced under a single nearly optimum environment. Also, with the complexity of response to early-season cold stress, observations on a number of features of the recombinant inbred progenies during seed production are provided from unreplicated plots to give information on possible traits that could affect the expression of cold tolerance. The parents and RILs used in this study exhibited a spread of only 2–4 days difference in flowering time. Seeds were fairly uniform in each inbred line in color and size. Seed color of inbred lines range from pearly white to plain white and no inbred progenies exhibited testa, as both parents lack testa. All inbred lines exhibited comparable seed size, as both parents have similar seed weight.

Evaluation of cold germinability and field emergence of RILs

The parents (RTx430 and P610727) used in the study are polymorphic for various traits on the basis of germinability at 30 and 12°C, field emergence and vigor (Table 1). The parents exhibited moderate to low seedling vigor based on the scale used by Maiti

Table 1 Summary statistics for early-season cold tolerance traits evaluated in the study. Values inside parentheses for parents and means represent standard error of the mean

Traits	Parents		RIL population				
	RTx430	PI610727	Mean	Range	Skewness	Kurtosis	H
%Germ@12°C	38.67 (3.27)	98.67 (1.63)	73.13 (1.10)	28–98%	−0.46	−0.25	0.80
%Germ@30°C	60.00 (5.66)	100.00 (0.00)	88.23 (0.79)	48–100%	−1.23	1.53	0.83
%Emergence_USDA	16.00	68.00	31.85 (1.28)	2–77%	0.73	−0.02	0.72
%Emergence_ND	9.33	68.67	34.15 (1.24)	4.47–79.33%	0.73	0.10	0.68
Vigor_USDA	3.88	2.50	3.03 (0.05)	1.25–4.25	−0.04	−0.04	0.65
Vigor_ND	4.00	2.33	3.29 (0.04)	1.83–4.25	−0.08	−0.64	0.62

H heritability estimate

et al. (1981). The soil and ambient air temperatures recorded daily during early-season planting from April 1 to April 30, 2009, are shown in Electronic Supplementary Material Fig. S1 to provide information on the conditions for cool season cultivation. The mean temperature during the early-season planting conducted in this study was cool for both soil and ambient air (14.9 and 17°C, respectively).

The RIL mean values for cold and optimal temperature germinability were 73.13 and 88.23%, respectively (Table 1). The mean field emergence values for the two locations, USDA_LBK and ND_TAMU, were 31.8 and 34.1% respectively, while the mean seedling vigor ratings were 3.0 and 3.3.

The values for heritability (*H*) estimated for each trait evaluated are shown in Table 1. *H* values of 0.82 and 0.80 were obtained for cold and optimum temperature germinability, respectively. The *H* values for field emergence and seedling vigor at two locations were slightly lower (Table 1).

The traits evaluated in this study displayed significant correlations with each other (Table 2). The data for optimum and cold temperature germinability, field emergence from USDA and ND displayed

significant positive correlations with each other. Lower positive correlations were found between cold/optimal temperature germinability data and field emergence at both locations. Seedling vigor was negatively correlated to germinability due to the decreasing trend of the scoring system applied (1—excellent vigor to 5—low vigor). However, vigor ratings between the two locations tested were significantly correlated with each other, with a value of 0.45 (Table 2).

The trend for field emergence trait exhibited by the recombinant inbred lines is shown in Fig. 1. The frequency distribution showed that most of the inbred progenies clustered towards the mid-parent value. However, a number of inbreds showed significantly higher values than the better parent PI610727 at both locations. This latter observation indicates that transgressive segregation for field emergence (during early-season planting) exists in this population.

Genetic map of sorghum

The linkage map constructed for the recombinant inbred population consists of 141 SSR markers and

Table 2 Correlation analysis of traits

	%Germ@30°C	%Germ@12°C	Field emerge USDA	Field emerge ND	Vigor USDA	Vigor ND
%Germ@30°C	—					
%Germ@12°C	0.74**	—				
Field emerge USDA	0.33*	0.37*	—			
Field emerge ND	0.31*	0.32*	0.74**	—		
Vigor USDA	−0.25*	−0.56**	−0.56**	−0.47**	—	
Vigor ND	−0.19 ns	−0.25*	−0.26*	−0.27*	0.45**	—

Statistical significance for each value is shown as: ** significant at $P = 0.01$; * significant at $P = 0.05$; ns not significant

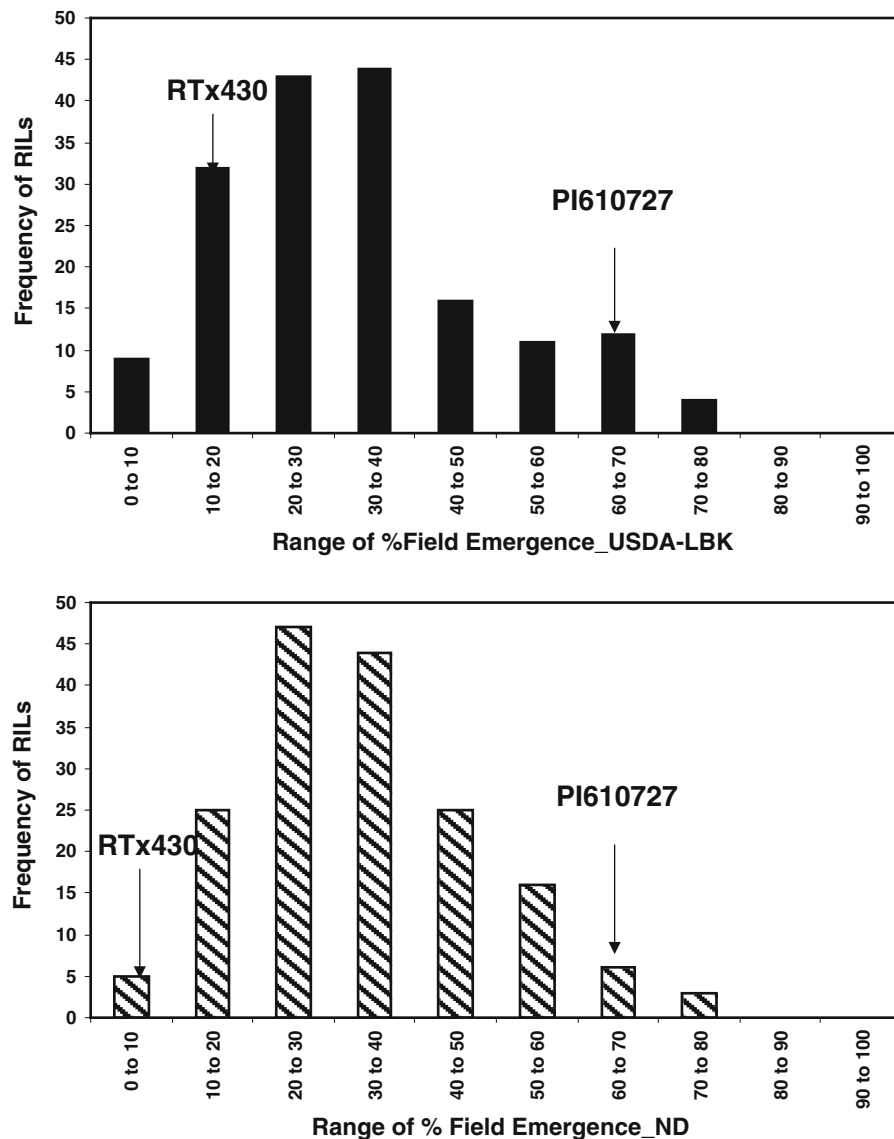


Fig. 1 Frequency distribution of means for %early field emergence (*Fearlygerm*) of the 171 RILs in two locations; Cropping Systems Research Laboratory farm, Lubbock, Texas (USDA_LBK) and Texas A&M Agrilife farm at New Deal,

Texas (ND_TAMU). The experiment was sown for early-season planting on April 1, 2009. Values for parents are indicated by the arrows

spans a total of 1005.1 cM (Supplementary Fig. S2). The map is composed of 10 linkage groups designated SBi1 to SBi10 based on chromosomal designation by Kim et al. (2005). In general, the positions of markers were consistent with previously reported maps for sorghum which are populated by SSR markers (Menz et al. 2002; Wu and Huang 2007; Salas Fernandez et al. 2008).

Identification of QTL for cold and optimal temperature germinability

QTL for cold and optimal temperature germinability traits were represented as *Germ12* and *Germ30*, respectively (Table 3, Fig. 2). QTL analysis revealed two and three QTL each for cold and optimal germinability, respectively, with significant LOD

Table 3 Information on QTL identified from the recombinant inbred population through the application of composite interval mapping (CIM)

Trait	QTL name	Location	Chr*	LOD score	QTL peak position (cM)	Interval (cM)	Additive effect	PVE (%)
%Germ@30°C	<i>Germ30-1.1</i>	–	1	2.5	8.0	5.0–15.5	1.92	4.5
	<i>Germ30-1.2</i>	–	1	3.0	81	80–121	–1.9	7.21
	<i>Germ30-2.1</i>	–	2	6.0	102	90–115	–3.70	14.5
%Germ@12°C	<i>Germ12-2.1</i>	–	2	9.0	99	90–115	–3.85	20.0
	<i>Germ12-9.2</i>	–	9	2.3	132	125–145	–2.03	3.2
Field emergence	<i>Fearlygerm-1.2</i>	USDA_LBK	1	2.3	125	120–132	1.88	3.68
	<i>Fearlygerm-7.1</i>	USDA_LBK	7	2.8	38	35–42	2.6	5.0
	<i>Fearlygerm-9.2</i>	USDA_LBK	9	4.8	133	125–145	–3.13	15.5
	<i>Fearlygerm-9.3</i>	USDA_LBK	9	3.8	154	145–165	–3.13	12.8
	<i>Fearlygerm-1.1</i>	ND_TAMU	1	2.8	8	0–16	1.57	3.12
	<i>Fearlygerm-4.1</i>	ND_TAMU	4	3.8	50	40–60	–3.11	8.5
	<i>Fearlygerm-9.1</i>	ND_TAMU	9	3.8	30	25–35	–1.55	2.56
	<i>Fearlygerm-9.3</i>	ND_TAMU	9	2.4	154	149–165	–2.68	8.8
Seedling vigor	<i>Fearlyvigor-4.1</i>	ND_TAMU	4	3.8	40	37–43	–2.11	10.5

Chr* refers to chromosomal location based on Kim et al. (2005)

values ranging from 2.3 to 9.0 (Table 3). The QTL *Germ12C-2.1* and *Germ12C-9.2* displayed negative additive effects with 20.0 and 3.2% PVE contributions, respectively, to the overall phenotypic variability of the population (Table 3). Meanwhile, QTL *Germ30-1.1* showed a positive additive effect and *Germ30-1.2* and *2.1* both showed negative additive effects. Notably, *Germ12-2.1* and *Germ30-2.1* coincide and co-localize with each other with relatively high contribution to phenotypic variation (PVE) (Fig. 2). The flanking markers for each QTL are shown in Table 4.

Identification of QTL for early-season field emergence and seedling vigor

The QTL for field emergence were represented as *Fearlygerm*(USDA_LBK) or as *Fearlygerm*(ND-TAMU), while seedling vigor was represented as *Fearlyvigor*(ND_TAMU) (Table 3, Fig. 2). Four QTL each were identified for field emergence from the USDA_LBK and ND_TAMU locations, with LOD values ranging from 2.3 to 4.8. The QTL for the USDA_LBK location were identified in chromosomes 1 (*Fearlygerm-1.2*), 7 (*Fearlygerm-7.1*), and 9 (*Fearlygerm-9.2* and *Fearlygerm-9.3*). Two of the QTL (*Fearlygerm-1.2* and *Fearlygerm-7.1*) showed

positive additive effects, which means that the alleles of markers in this region were from RTx430. For *Fearlygerm-9.2* and *Fearlygerm-9.3* which showed negative additive effects, specifically the alleles of markers in this region were derived from the PI610727 parent. The QTL for the ND_TAMU location were found in chromosomes 1 (*Fearlygerm-1.1*), 4 (*Fearlygerm-4.1*), and 9 (*Fearlygerm-9.1* and *Fearlygerm-9.3*). Only *Fearlygerm-1.1* exhibited positive additive effect while the rest of QTL had negative additive effects, which means that these QTL were contributed by PI610727. One QTL (*Fearlygerm-9.3*) was identified in both locations which indicates its stability in influencing the expression of field emergence. The markers nearest to the peak of each QTL for field emergence in each location are shown in Table 4. Only one QTL for seedling vigor, *Fearlyvigor-4.1*, was identified as significant by CIM (Table 3). This QTL was found in the ND_TAMU location only.

Correspondence of QTL for early-season cold tolerance identified in this study with previously reported studies

An alignment of the QTL map derived in this study was performed specifically in comparison with the

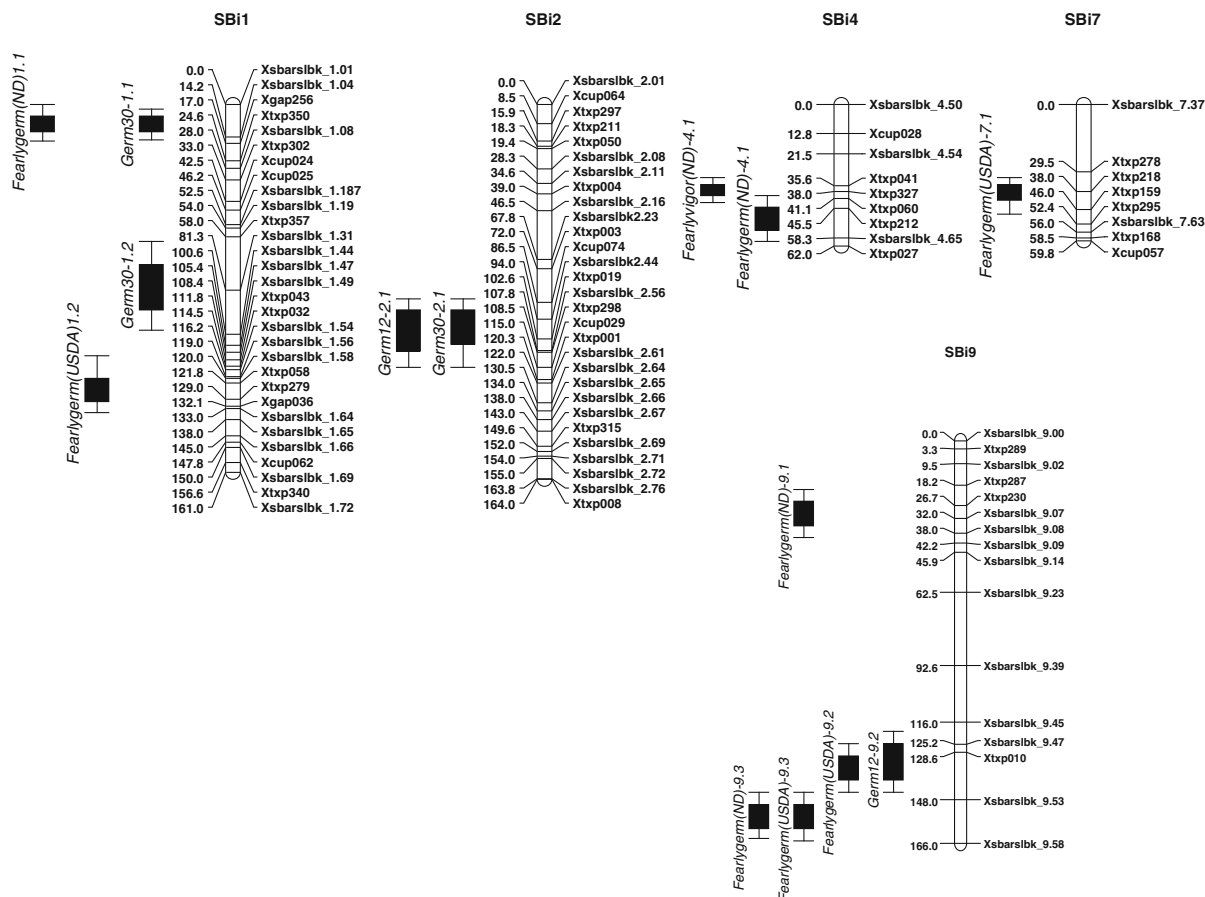


Fig. 2 Molecular linkage map of specific sorghum chromosomes harboring QTL for early-season cold tolerance. The QTL are shown as *solid bars* on the left side of each linkage

group. *Extended lines* represent 2-LOD interval for each QTL. Values on *left side* of each linkage group refer to estimated position of markers in cM units

previous genetic map reported by Knoll and Ejeta (2008) and Knoll et al. 2008 (Fig. 3). The two studies (Knoll and Ejeta 2008 and this study) differ substantially in the type of markers populating each map, as the former study primarily employed randomly amplified polymorphic DNA (RAPD). Likewise, the orientation of Sbi1 presented by Knoll et al. (2008) has to be flipped 180° to correspond to the known orientation of the chromosome (Fig. 3). However, some correspondences were observed based on common SSR markers between the two studies. For chromosome 1 (Sbi1), correspondence between QTL regions for germinability and field emergence to that of late emergence, anchored by Xtxp043 (shown by solid line in Fig. 3), were observed in the aligned genetic maps.

Discussion

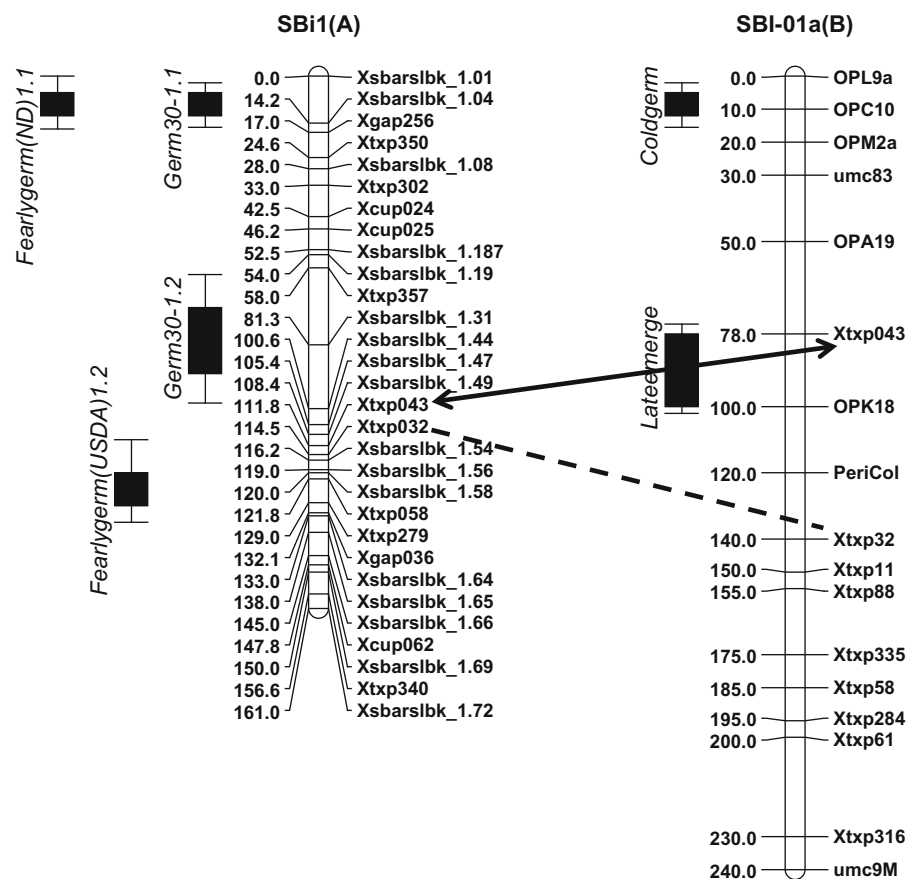
Tolerance to early-season cold temperature is an important and highly desirable trait for warm-season cereal crops like sorghum and rice. It is important particularly for sorghum which requires excellent initial crop stand during early-season planting to ensure stable yield. Early-season cold tolerance has also been proposed as one of the traits that can confer higher efficiency to sorghum in its use of soil moisture from spring precipitation and withstand cool soils associated with minimum tillage (Bacon et al. 1986). However, breeding for early-season cold tolerance has been challenging due to strong environmental effects, and it has been shown that recurrent selection using adapted/elite populations

Table 4 Information on flanking markers associated with QTL regions identified by CIM

Trait	QTL name	Location	Chr*	LOD score	QTL peak position (cM)	Flanking markers
%Germ@30°C	<i>Germ30-1.1</i>	–	1	2.5	8.0	<i>Xsbarslbk 1.04</i>
	<i>Germ30-1.2</i>	–	1	3.0	81	<i>Xtxp043</i>
	<i>Germ30-2.1</i>	–	2	6.0	102	<i>Xsbarslbk 2.56</i>
%Germ@12°C	<i>Germ12-2.1</i>	–	2	9.0	99	<i>Xsbarslbk 2.56</i>
	<i>Germ12-9.2</i>	–	9	2.3	132	<i>Xtxp010</i>
Field emergence	<i>Fearlygerm-1.2</i>	USDA_LBK	1	2.3	125	<i>Xtxp58</i>
	<i>Fearlygerm-7.1</i>	USDA_LBK	7	2.8	38	<i>Xtxp218</i>
	<i>Fearlygerm-9.2</i>	USDA_LBK	9	4.8	133	<i>Xtxp010</i>
	<i>Fearlygerm-9.3</i>	USDA_LBK	9	2.1	154	<i>Xsbarslbk9.58</i>
	<i>Fearlygerm-1.1</i>	ND_TAMU	1	2.8	8	<i>Xsbarslbk1.04</i>
	<i>Fearlygerm-4.1</i>	ND_TAMU	4	3.8	50	<i>Xtxp212</i>
	<i>Fearlygerm-9.1</i>	ND_TAMU	9	3.8	30	<i>Xsbarslbk9.07</i>
	<i>Fearlygerm-9.3</i>	ND_TAMU	9	2.4	154	<i>Xsbarslbk9.58</i>
Seedling vigor	<i>Fearlyvigor-4.1</i>	ND_TAMU	4	3.8	40	<i>Xtxp060</i>

Chr* refers to chromosomal location based on Kim et al. (2005)

Fig. 3 Alignment of QTL map of sorghum chromosome 1 from this study SBI1(A) and from a previous genetic map SBI-01a(B) reported by Knoll et al. (2008). Note that the map of Sbi-01a(B) from Knoll et al. (2008) was flipped from its original orientation to facilitate alignment. *Solid line* connecting markers represents shared markers that are associated with QTL, while *dashed line* represents common marker between the two maps. Values on *left side* of each linkage group refer to estimated position of markers in cM units



resulted in only a small improvement of the trait (Bacon et al. 1986). Thus efforts to search for new sources of cold tolerance, followed by their effective and efficient incorporation into sorghum parental lines, has received considerable attention (Yu and Tuinstra 2001; Cisse and Ejeta 2003; Franks et al. 2006).

In this report, genetic dissection of tolerance to early-season cold tolerance was performed through the establishment of a genetic map using SSR markers and subsequent QTL analysis from the population developed from RTx430 × PI610727. The map obtained from this study was consistent with previously published genetic maps and the sorghum sequence (www.phytozome.net; www.gramene.org). Our current map distance (1005.1 cM for all ten chromosomes) was comparable to that reported by Wu and Huang (2007) at 997.5 cM and by Salas Fernandez et al. (2008) at 1357.7 cM. The lower total map distances reported in these studies compared to the high-density map from Menz et al. (2002) can be attributed to limited coverage of the genomic regions for a number of chromosomes, specifically SBI 5 and 8 in our study, where fewer polymorphisms for the SSR set employed are found in these chromosomes.

The heritability values for all traits studied were relatively high (0.62–0.83), which suggest this new parental line could be used for breeding for early-season cold tolerance trait. A significant and positive correlation between optimum temperature (30°C) and cold (12°C) germinability was found among the inbred progenies (Table 2), which indicates that vigorous germination is one of the important components of cold germinability. This strong relationship between the two traits has also been shown for rice (Fujino et al. 2004) and for sorghum using a different population (Knoll and Ejeta 2008). However, the relatively lower but significant positive correlation between cold germinability and actual field emergence data (Table 2) indicates that controlled testing for germinability could serve as one of the indicators for cold tolerance to ensure that seeds are capable of germination, but that field emergence at early-season planting must be the main screening method to ensure efficient selection.

In the present study, QTL analysis using the CIM module was applied to dissect the genetic basis of early-season cold tolerance in sorghum. In general, the CIM method employed to confirm results from single

marker analysis increased the detection power and provided an estimate of an experiment-wise threshold value which minimized type I error (Basten et al. 2003).

A total of 14 QTL associated with four traits for cold tolerance was detected in this study (Table 3, Fig. 2). The QTL were located in five sorghum chromosomes, namely 1, 2, 4, 7, and 9. In particular, it was shown that chromosome 9 harbors four QTL for field emergence. Three of these four QTL, *Fearlygerm(USDA)-9.2*, *Fearlygerm(USDA)-9.3*, and *Fearlygerm(ND)-9.3* displayed medium to strong effects (8.8, 12.8, and 15.5%, respectively) on overall phenotypic variance. The additive effects of these QTL were negative which indicated that the allele contributing to the observed increase in field emergence was inherited from PI610727. Further analysis showed that two of these QTL, *Fearlygerm(USDA)-9.3* and *Fearlygerm(ND)-9.3* co-localized with each other, had the same flanking marker and spanned the same genome region in chromosome 9. Moreover, the QTL *Fearlygerm -9.3* explained a large proportion (15.5%) of the phenotypic variance and could provide useful markers for selecting field emergence (Table 3). An informative marker for this QTL was identified as *Xsbarslkb9.58* (Table 4).

The QTL for cold and optimal temperature germinability detected in chromosome 2, *Germ 30-2.1* and *Germ 12-2.1*, were found to co-localize under the same genome region and had the same flanking marker, *Xsbarslkb2.56* (Fig. 2, Table 4). These QTL also explained a large proportion of the phenotypic variance for each trait and could exert strong effect on both traits. This result provides genetic support to the findings that cold and optimal temperature germinability are strongly correlated with each other and could be under the control of a similar set of genes. Similarly, the co-localization between *Germ12-9.2* and *Fearlygerm(USDA)-9.2* and that of *Germ30-1.1* and *Fearlygerm(ND)-1.1* in chromosome 9, were found significant of interest. These QTL regions provide genetic support to the observation that cold germinability under controlled and optimum conditions are important traits that contribute to field emergence in the early-season planting. Only one QTL, *Fearlyvigor-4.1*, was detected for early-season vigor in the present population. This QTL exerts a negative additive effect which means that it is contributed by an allele from the PI610727 parent.

Notably, the QTL regions identified in this study have been reported to harbor QTL for other traits including grain quality and resistance to grain mould (Rami et al. 1998; Klein et al. 2001). The QTL for early vigor and early germination in chromosome 4 have also been identified to harbor QTL for variation in floury endosperm trait (Rami et al. 1998). Meanwhile, some QTL have been identified for resistance to grain mould in the region of the cluster of QTL in the distal end of chromosome 9 (Klein et al. 2001). These findings could be coincidental, or they may indicate that the genes underlying variation in these phenotypes are also associated with early vigor. Closer examination of grain quality attributes may shed further light on the physiological basis of cold tolerance and may be important for the implementation of marker-assisted selection for this trait in applied breeding programs.

In this report, QTL mapping for four traits associated with tolerance to early-season cold tolerance in sorghum was performed using an RIL population. Here we tagged the genome regions that have significant contributions to traits for early-season cold tolerance using a new source of cold tolerance, PI610727. Identification of new QTL that are stable across environments and those that control several correlated traits could provide new information on DNA markers that could be highly informative and effective in marker-assisted selection targeting early-season cold tolerance. However, it is proposed that further studies on QTL analysis for early-season cold tolerance in sorghum are warranted, specifically towards utilization of a combination of markers in introgression for enhanced field emergence and vigor at early-season planting.

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